



Week 7 Lecture 5

▼ Class	BSCCS2003
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🔗 Materials	
# Module #	42
▼ Type	Lecture
# Week #	7

Scaling

Replication and Redundancy

- Redundancy:
 - Multiple copies of same data
 - Often used in connection with backups - even if one fails, others survive
 - One copy is still the master
- Replication:
 - Usually in context of performance
 - May not be for purpose of backup
 - Multiple sources of same data - less chance of server overload
 - Live replication requires careful design

BASE vs ACID

- "Basically Available", "Soft state", "Eventually consistent"
 - Winner of worst acronym award, lol
- Eventual consistency instead of Consistency
 - Replicas can take time to reach a consistent state
- Stress on high availability of data

Replication in traditional DBs

- RDBMS replication is possible
- Usually server cluster in the same data center
 - Load balancing
- Geographically distributed replication harder
 - Latency constraints for consistency

Scale-up vs Scale-out

- **Scale-up:** Traditional approach
 - Larger machine
 - More RAM
 - Faster network, processor
 - Requires machine restart with each scale change
- **Scale-out:**
 - Multiple servers
 - Harder to enforce consistency etc. → better suited for NoSQL/non-ACID
 - Better suited to cloud model: Google, AWS etc. to provide automatic scale-out, cannot do auto scale-up

Application Specific

- Financial transactions:
 - Cannot afford even slightest inconsistency
 - Only scale-up possible
- Typical web application
 - Social networks, media: Eventual consistency is OK
 - e-Commerce: only the financial part needs to go to ACID